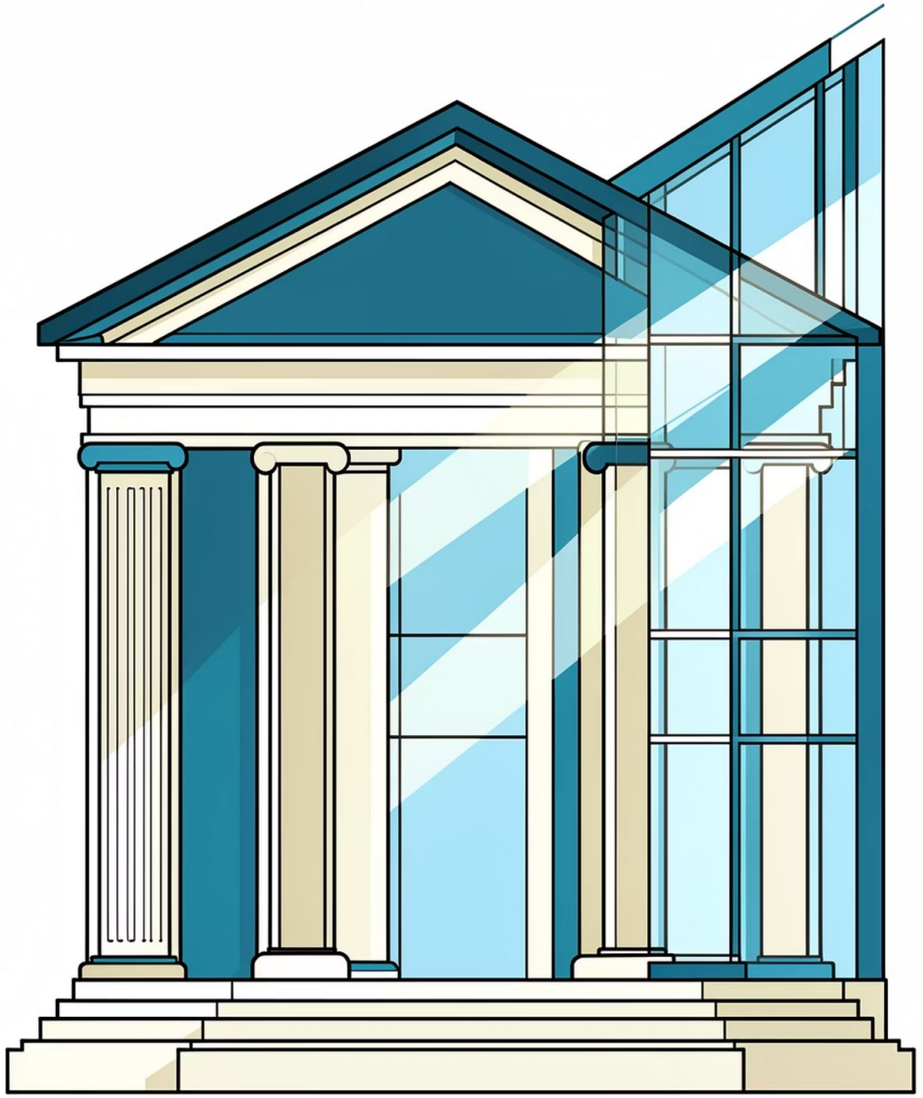
Legal Question Answering System using RAG

A Retrieval-Augmented Generation approach to making legal information accessible.



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The Idea

The Challenge

Legal texts are dense, technical, and difficult for the general public to interpret. Accessing reliable, plain-language explanations is a genuine barrier to legal literacy.

The Solution

An AI assistant that retrieves relevant legal texts and explains them in clear, simple language — making legal knowledge accessible to everyone.

Project Objective

Build a functional RAG system that answers legal questions by combining semantic search with a language model.

Semantic Retrieval

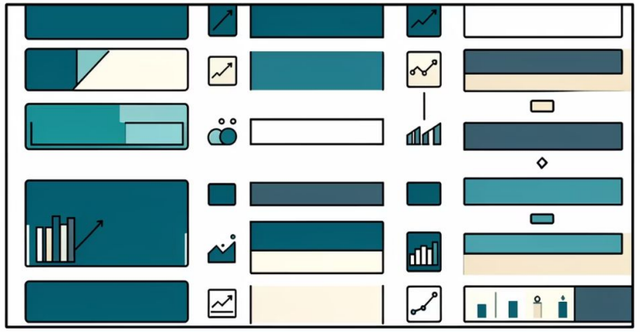
ChromaDB stores and retrieves the most relevant legal document chunks based on meaning, not just keywords.

Language Model

A Hugging Face model (flan-t5) generates clear, contextually grounded answers from retrieved passages.

Reliable Context

Answers are always grounded in retrieved source documents — reducing hallucination and improving trustworthiness.



DATASET

The Dataset

Dataset source : Kaggle

The dataset includes original legal texts and their simplified explanations.

[Indian Law & Supreme Court Database 2026 (NLP/RAG)]

Why It Fits RAG

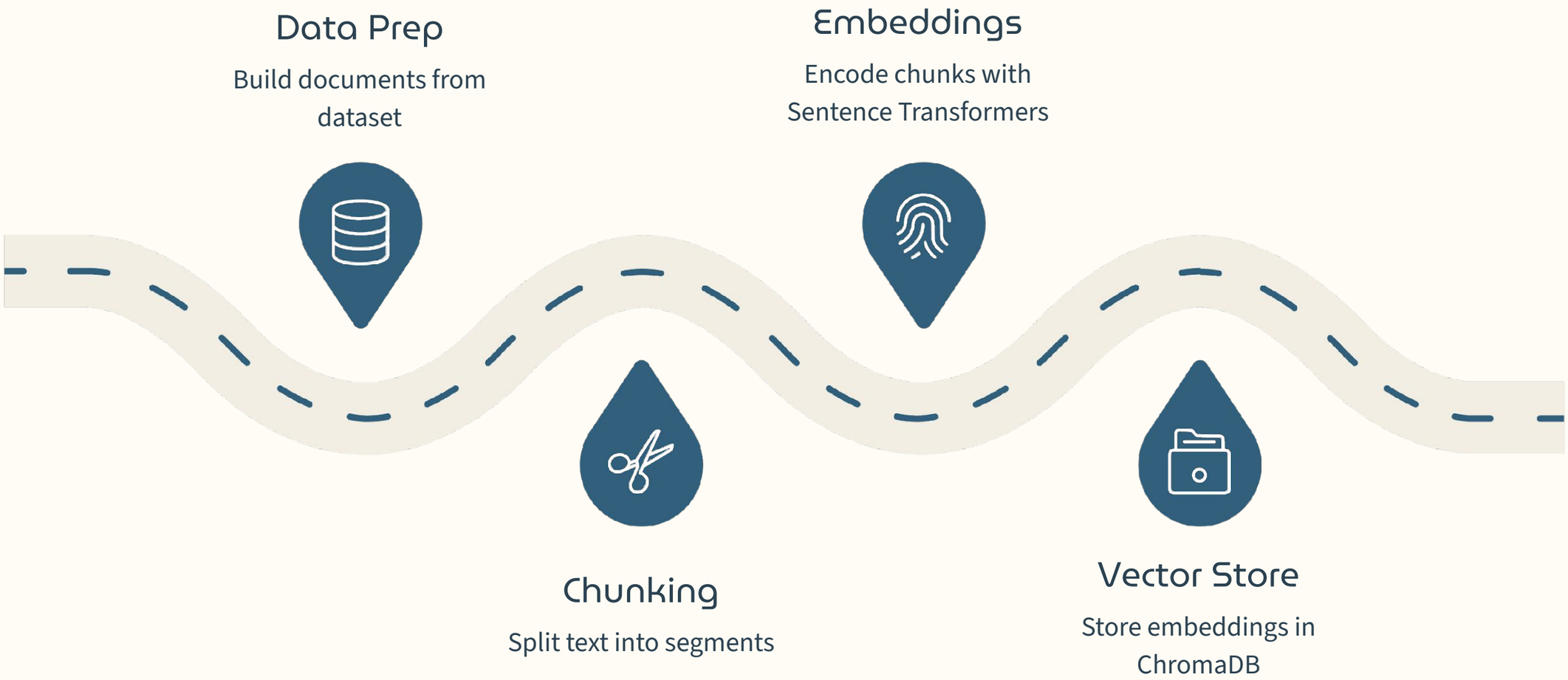
Each text chunk can be used on its own, and the simplified explanations help check if the answers are correct.

Two types of Content

Original text and simplified text, which help the model give accurate and easy-to-understand answers.

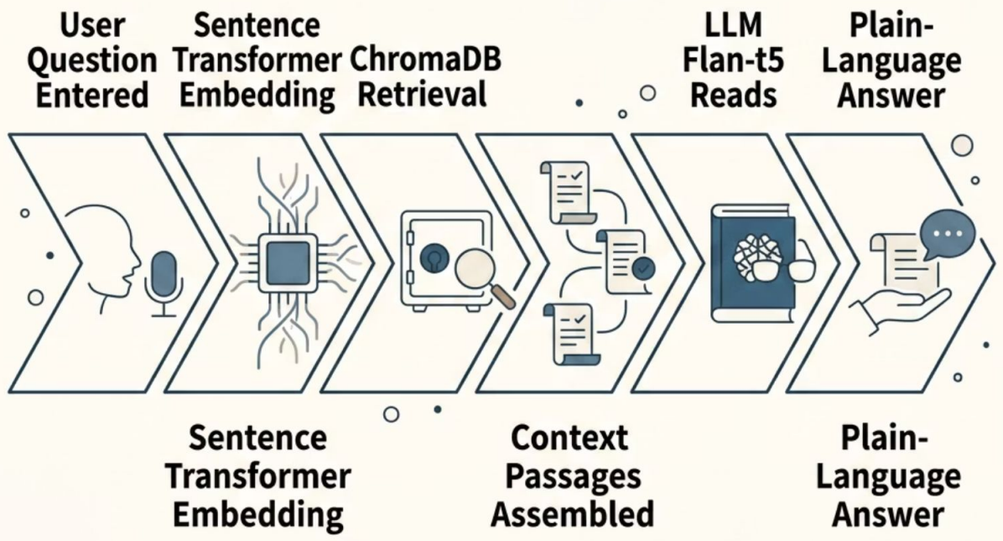
METHODOLOGY

The RAG Pipeline



The process clearly divides data preparation, retrieval, and answer generation to keep things simple and organized.

System Architecture



How It Works

A user asks a legal question. It is converted into a vector, matched with stored texts, and the most relevant ones are sent with the question to the model to generate an answer.

> The retrieval step ensures the model only "sees" relevant legal context, not the entire corpus.

Results & Demo

Example Question

"What are the rights of an accused person during a criminal trial in India?"

System Answer

The system retrieves relevant legal passages about Fundamental Rights, then generates a clear and simple explanation — explaining that these rights mainly protect individuals against the State and ensure basic freedoms and equality.

EVALUATION & EXPERIMENTS

What We Measured

Model & Prompt Experiments

- **flan-t5-base vs. large**: similar quality, large model notably slower
- **Basic prompt**: concise, direct answers
- **Simplified prompt**: more accessible language
- **Strict prompt**: longer, more cautious responses

Limitations & Difficulties

Known Limitations

Small Dataset

Limited diversity in training documents reduces answer variety.

No Deep Reasoning

the model cannot handle multi-step legal thinking.

Challenges Encountered

- Selecting a dataset that was both legal and RAG-compatible
- Handling repetitive outputs caused by similar retrieved chunks
- Understanding whether errors come from retrieval or from the model

Improvements & Conclusion

Future Improvements

01

Richer Data

Larger, more diverse legal datasets for broader coverage

02

Stronger LLMs

More powerful models for deeper, more nuanced responses

03

Better Retrieval

Improve how results are selected by sorting them better, removing irrelevant ones, and combining different search methods.

04

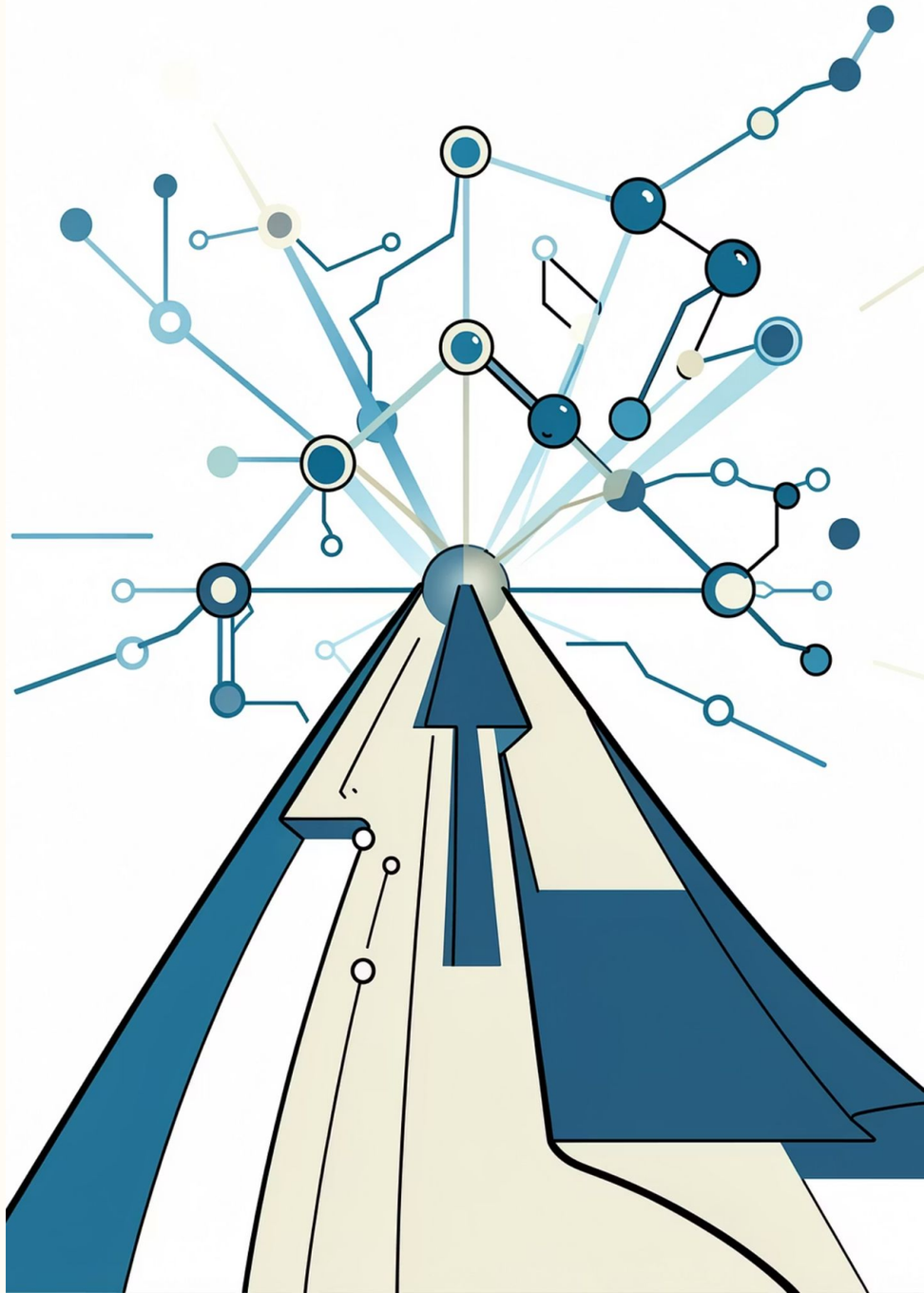
Memory

Keeping context from previous questions in a conversation.

Key Takeaways

RAG uses real documents to answer questions, which reduces incorrect or made-up answers compared to a model used alone.

The system demonstrates that AI can genuinely simplify complex legal information. **Data quality remains the most critical factor** — the pipeline is only as strong as the documents it retrieves from.



Thank You !